

Phantom Noise

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Overview

We scanned a NEMA IQ phantom back to back on the DMI and Omni. The phantom was filled with a 4:1 sphere to background ratio. The acquisition was 10 minutes long on each scanner and then reconstructed as shorter time-points.

The Omni was reconstructed with OSEM, QClear as well as the machine learning options (MPDL and HPDL). The DMI has ToF and QClear reconstructions. The Omni scans were reconstructed at multiple durations (60, 90, 120, 150, 180 seconds), the DMI is only 150 seconds, the current clinical default. The Omni scans were also replicated multiple times for more robust statistics, that process has not yet been done on the DMI data.

ROIs

ROIs were drawn on the CT, then a MIM workflow was used to copy them to the various PETs and extract SUV metrics.

1. Six spherical ROIs drawn inside each sphere
2. Two cylindrical ROIs in the lung insert, one near the spheres the other near the background
3. Fifty circular ROIs in the background, 10 on 5 different slices.

Noise Metrics

Noise (also referred to here as background variance) is the Coefficient of Variation (CoV) in the 50 background ROIs. We also evaluate Image Roughness, which is the mean of the CoV for each ROI. Both metrics are from work by Shan Tong (Tong et al. 2010)

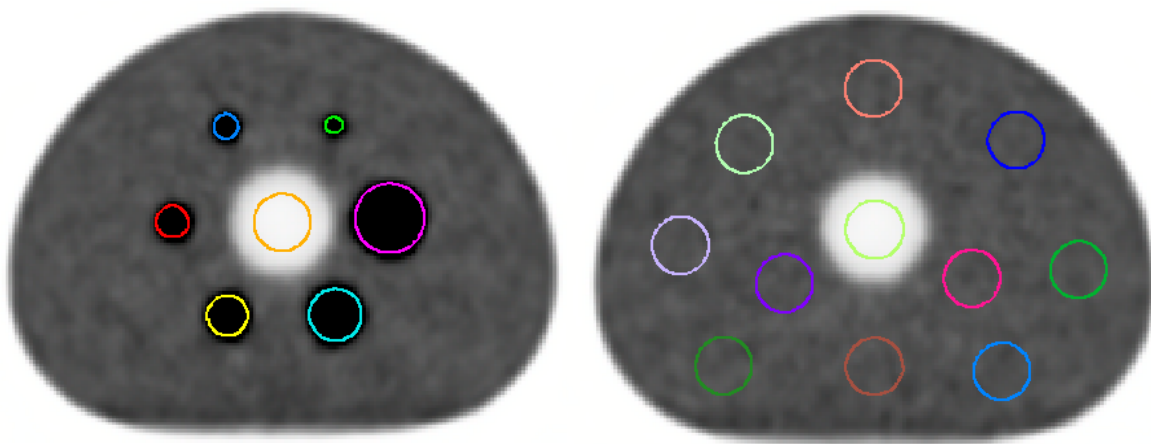
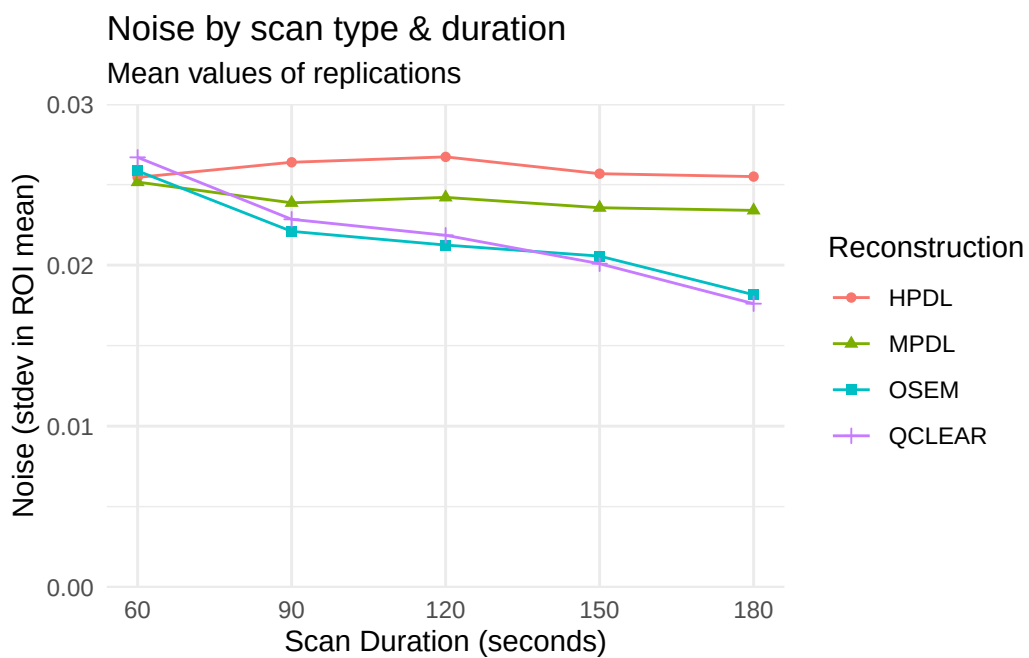


Figure 1: ROI Positions

Omni Noise Plots



Noise by scan type & duration

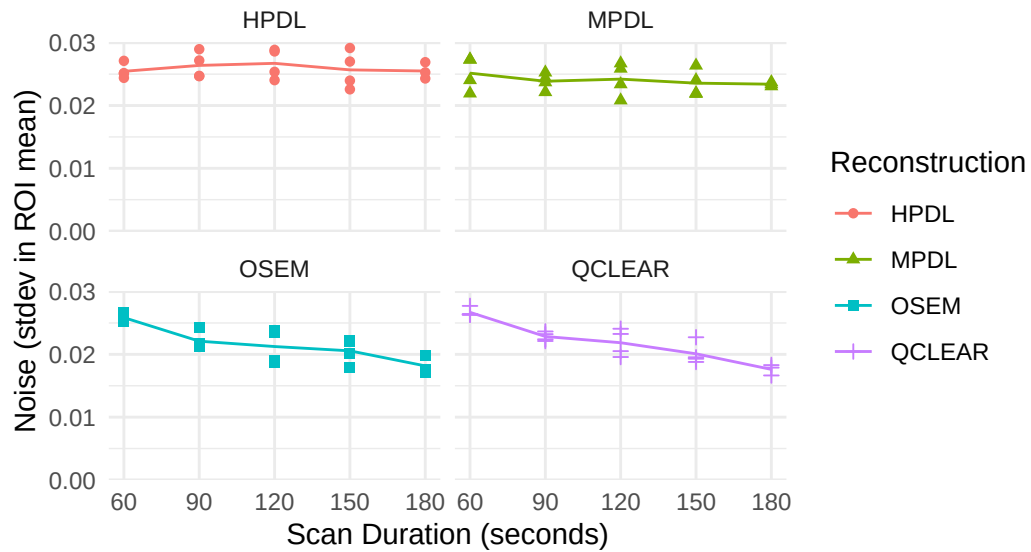


Image Roughness in ROI group by scan type/duration

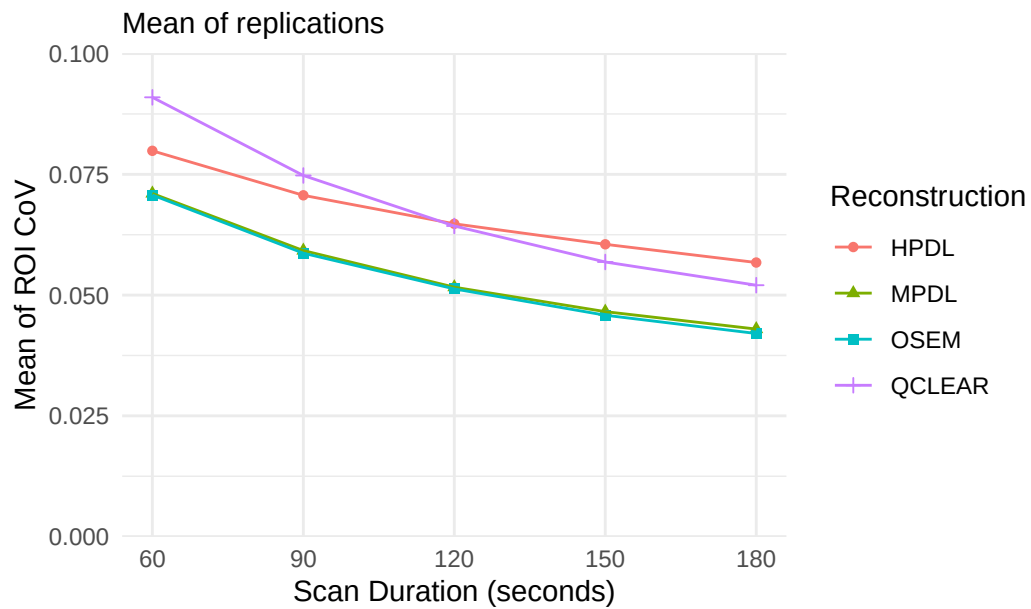
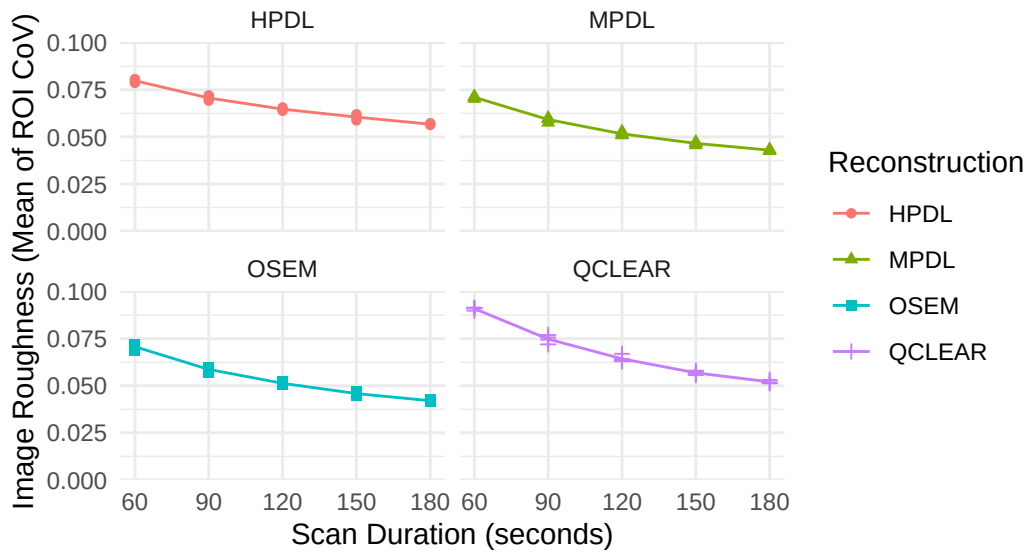
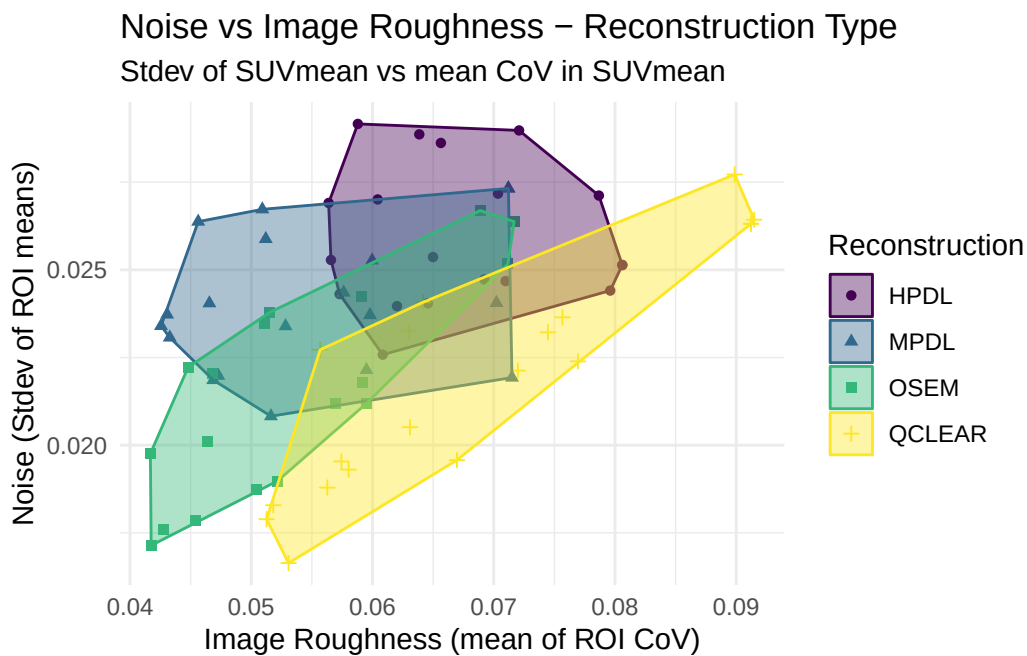
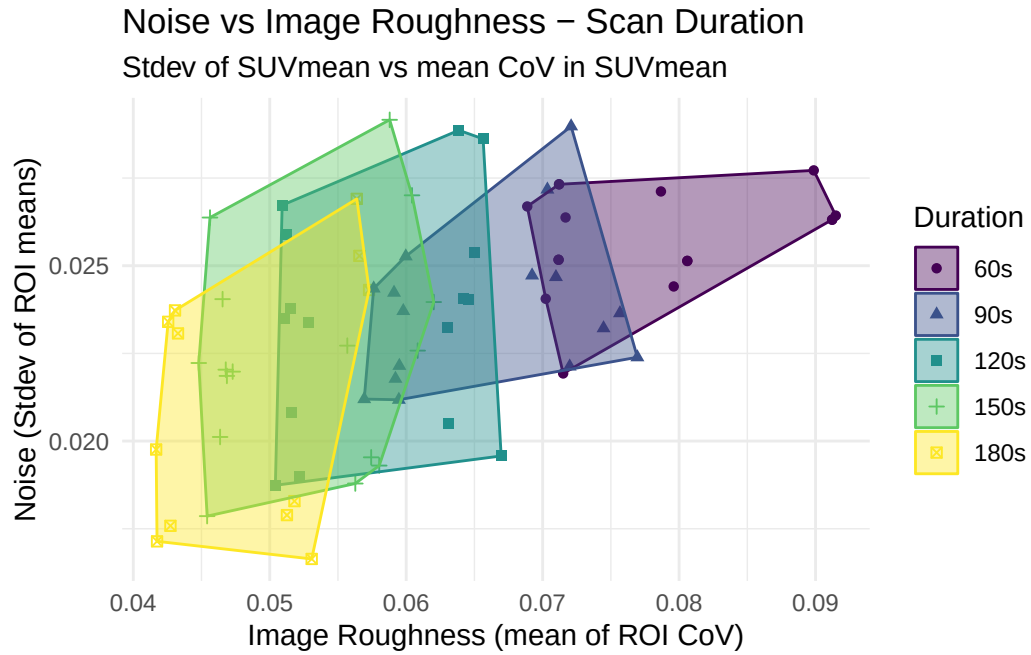


Image Roughness by scan type & duration



Comparison between image roughness and noise





Tong, S, A M Alessio, and P E Kinahan. 2010. “Noise and Signal Properties in PSF-Based Fully 3D PET Image Reconstruction: An Experimental Evaluation.” *Physics in Medicine and Biology* 55 (5): 1453–73. <https://doi.org/10.1088/0031-9155/55/5/013>.